JavaScript

Fetch API & Async/Await

Where Does This Data Come From?

Every app you use – Instagram, Zomato, Google Maps – shows you fresh data.
But where does it come from?
It comes from the internet, through something called an API!

The Internet is Like a Tea Stall

You Go to the tea stall and say, "One cutting chai, please!"
The chaiwala makes it and hands it to you.

The internet works the same way:
You (the browser) ask a server for data.
The server prepares it and sends it back!

APIs Are Like IRCTC Train Tickets

When you book a train ticket on IRCTC:

1. You fill a form (your request)
2. IRCTC checks availability (the server processes)
3. You get your ticket (the response)

An API is the middleman between you and the server.
fetch() is how JavaScript books that ticket!

Async Means Don't Freeze, Just Wait

Imagine you order food on Zomato.
Do you stand at the door and freeze until it arrives? No!
You keep doing other things – watching TV, scrolling reels.

That's what async means in JavaScript:
Don't stop everything. Keep working while you wait for data.

Mumbai Local Train is Async Too!

At a Mumbai local station, many trains come and go. Passengers don't wait for one train to finish its journey before the next one departs.

JavaScript is the same – it can send multiple fetch requests without waiting for each one to finish first!

fetch() is Like Sending a Peon

In an office, you send a peon to get a file from another department.
You don't go yourself – you send someone and keep working.

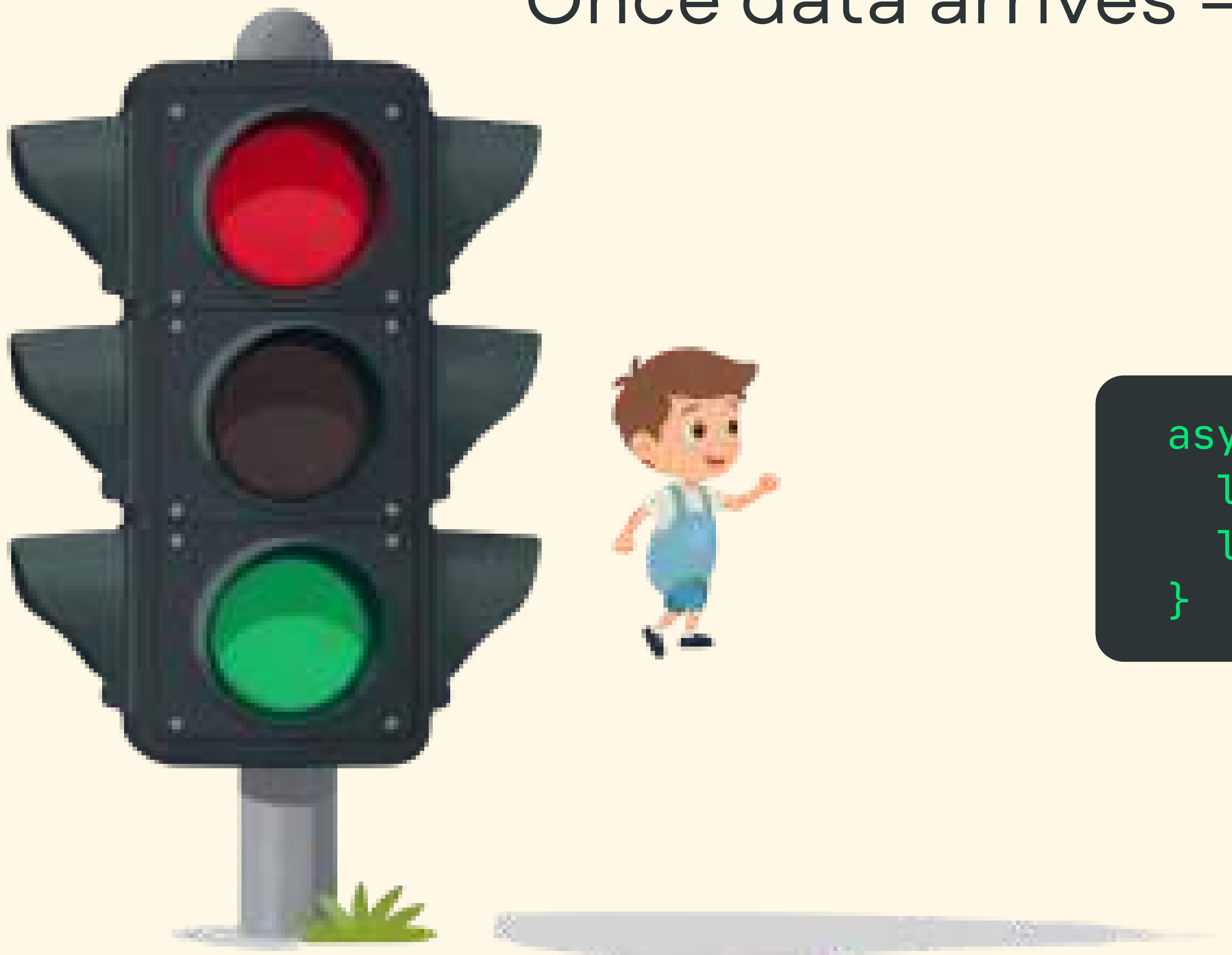
fetch() does the same thing:
It sends a request to a URL and brings back data for you.

```
fetch('https://api.example.com/data')
```

async/await Are Like Traffic Signals

At a Bangalore traffic signal, when it's red – you stop and wait.
When it turns green – you go!

async/await works the same way:
await = red signal (pause here, wait for data)
Once data arrives = green signal (continue running code)



```
async function getData() {
  let response = await fetch(url);
  let data = await response.json();
}
```

JSON - The Neatly Packed Mumbai Dabba

A Mumbai dabbawala delivers lunch in a tiffin box.
Each compartment has something different – rice, dal, sabzi.

JSON is like that tiffin box:
Data is neatly packed in labeled compartments.
Each piece has a name (key) and a value.

```
{
  "name": "Rahul",
  "age": 14,
  "city": "Mumbai"
}
```

Live Code Demo - Magic Data Fetcher

Let's fetch real data from the internet!
We'll use a free API that gives us random user info.

Here's the pattern:

```
async function getData() {
  let response = await fetch(
    'https://randomuser.me/api/'
  );
  let data = await response.json();
  console.log(data);
}
```

**This function sends a peon (fetch) to get data,
waits patiently (await), then unpacks the dabba (json)!**

Live Code Demo (Part 2)

Now let's add a button and show the data on screen!
We use innerHTML to display results in our HTML page.

```
async function getData() {
  let response = await fetch(
    'https://randomuser.me/api/'
  );
  let data = await response.json();
  let user = data.results[0];
  document.getElementById('output').innerHTML
    = user.name.first + ' ' + user.name.last;
}
```

We grab the first result and show the user's name on screen!

Result on Screen - Shabash!

When you click the button, real data appears!
You just fetched live information from the internet.

Shabash! You did it!

Output:
John Smith

Displaying Data with innerHTML

Think of innerHTML like writing on a chalkboard.
Your webpage is the board. JavaScript is the chalk.

document.getElementById('output').innerHTML = data.name;

This line finds a spot on the page and writes the fetched data there.
Just like a teacher writes the answer on the board!

```
document.getElementById('output').innerHTML
  = data.name + ' from ' + data.address.city;
```

Quick Quiz Time!

Let's see how much you remember!

5 quick questions coming up.
Think before you answer!

Quiz Questions (Part 1)

Q1: What does fetch() do?
A: It asks the internet for data – like sending a peon to get info.

Q2: What is async/await like in real life?
A: Like ordering food on Zomato and waiting for delivery.

Q3: What does the async keyword do?
A: It marks a function as one that will do some waiting inside.

Quiz Questions (Part 2)

Q4: What does await do?
A: It pauses the function until the data arrives – like a red signal.

Q5: What is JSON like?
A: A neatly packed Mumbai dabba – data organized in labeled compartments.

Wrap-Up & Homework

Today you learned:
- fetch() sends a request to the internet
- async/await helps you wait for data without freezing
- JSON organizes data neatly like a dabba
- innerHTML displays data on the page

Homework:
Change /users/1 to /users/2 or /users/3 in your code.
See what different data you get!

Keep coding – you're doing amazing!

Thank You!

Keep exploring the magic of JavaScript!

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